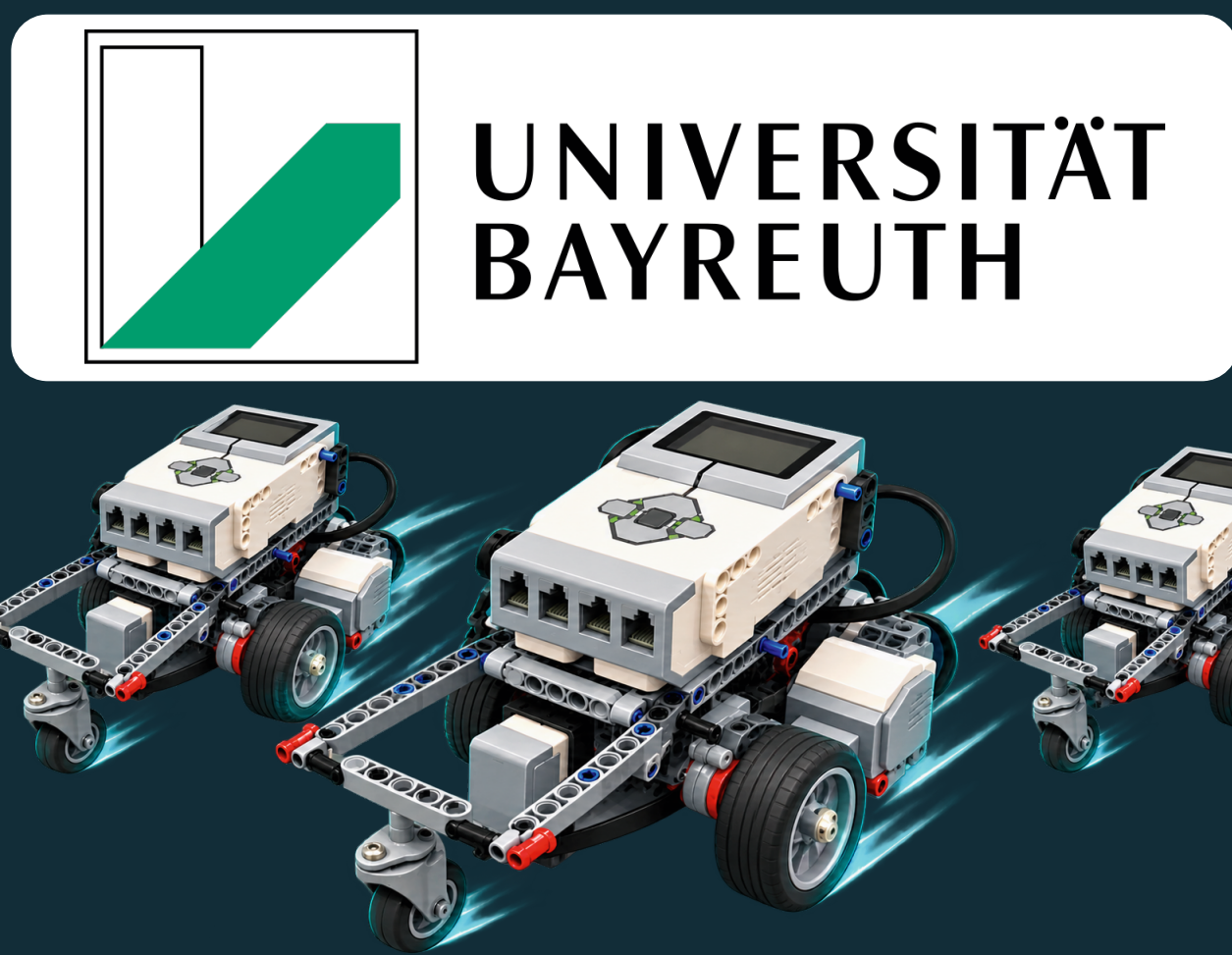


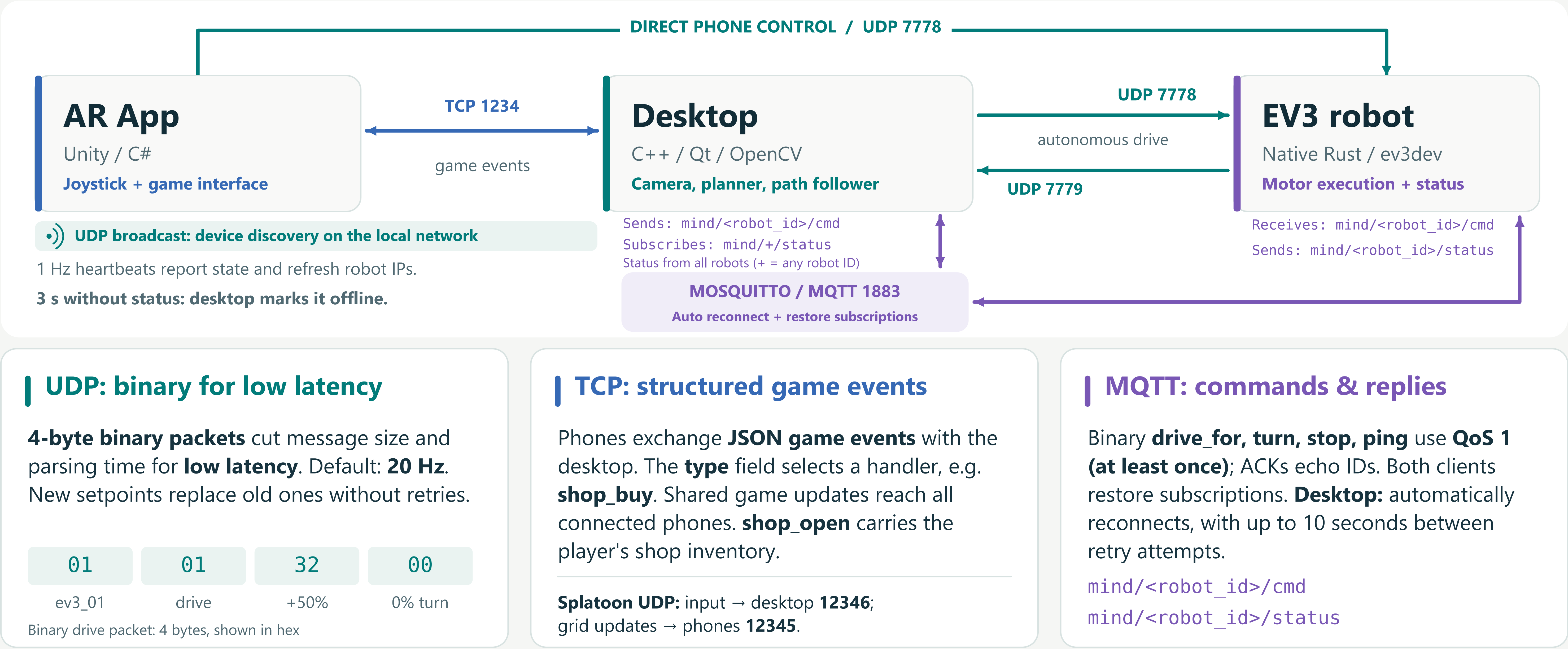


Robot Control & Communication

Collision avoidance, steering & network protocols

01 Network architecture

ARCHITECTURE



02 From a target to controlled movement

DESKTOP CONTROL

Plan around other robots

Global board routes + local collision avoidance

ROUTE

YIELD

CLEAR A NODE

1 / Route. Dijkstra selects the board-node route. Grid A* detours around parked robots. Clearance-checked corner rounding smooths the waypoints.

2 / Yield. A central coordinator watches nearby paths. The higher robot ID yields, replans around the other robot, or holds if no detour exists.

3 / Clear. If a destination is occupied, the blocker is routed to a free parking spot off the lanes. The arriving robot waits for the node to clear.

Follow the path, correct the heading

Camera pose → lookahead target → steering → wheels

planned path

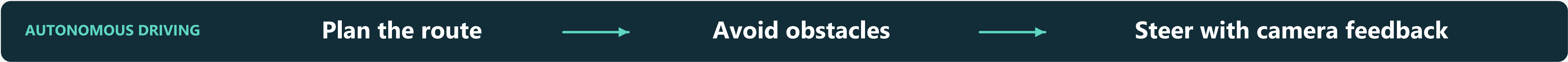
20 cm lookahead

heading error

1 / Observe. ArUco markers provide position and heading. A board homography converts camera pixels into physical centimetres.

2 / Steer. A lookahead point sets the heading error. A PID controller computes turn rate. Forward speed decreases in corners and near the goal.

3 / Settle. Sharp turns use pivot pulses and pauses for camera feedback to catch up. Calibration scales compensate drive and turn bias.



03 Inside the brick

RUST + EV3DEV

Ferris at the wheel

Four workers, one motor controller

Rust runs on the EV3's Linux system. A mutex protects the latest speed and turn command. Valid UDP packets wake the motor worker. A 10 ms timeout keeps watchdog checks running.

UDP
Decode + ID filter
Wake the motor loop

MOTOR
10 ms check interval
Regulated B/C motors

MQTT
Dispatch commands
Interruptible moves

HEARTBEAT
1 Hz to desktop
State + battery + ID

Motion commands: Speed and turn commands control continuous driving. Distance and angle moves run from a queue. The robot acknowledges each completed move.

Differential drive · B/C motors · 56 mm wheels

$$v_L = v - \frac{\omega b}{2} \quad v_R = v + \frac{\omega b}{2}$$

v: forward speed; ω: turn rate; b: wheel spacing (126 mm).
At the limit, both wheels slow proportionally to keep the curve.

When the robot stops

Timeouts measure time since the last valid input.

150 ms without a fresh drive packet
EV3 watchdog stops continuous driving.

400 ms without a camera update
Desktop commands zero speed and turn.

MQTT STOP command
Stops continuous driving and interrupts an active precision move.

MORE ABOUT THE PROJECT

Scan for project documentation